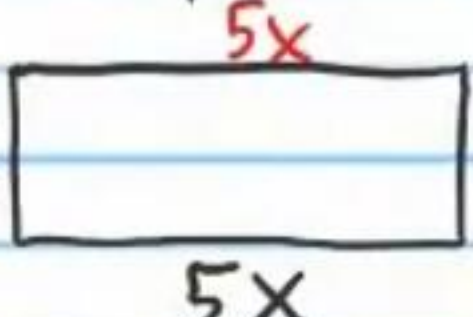
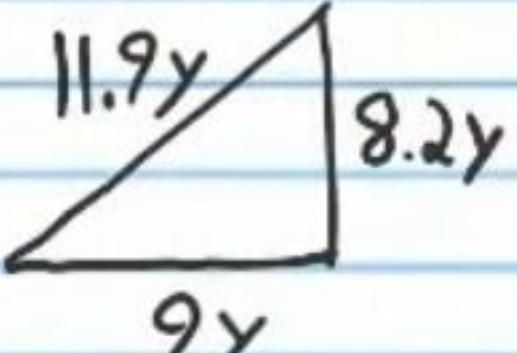


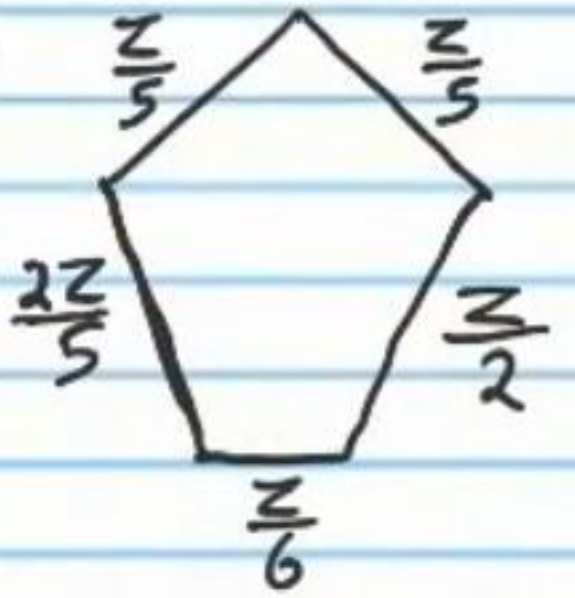
Applications of Algebra (Part I): Simplifying and Evaluating Algebraic Expressions

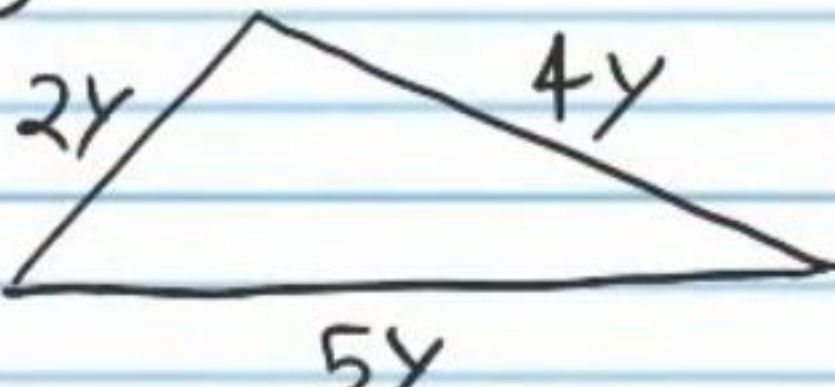
Simplifying Algebraic Expressions

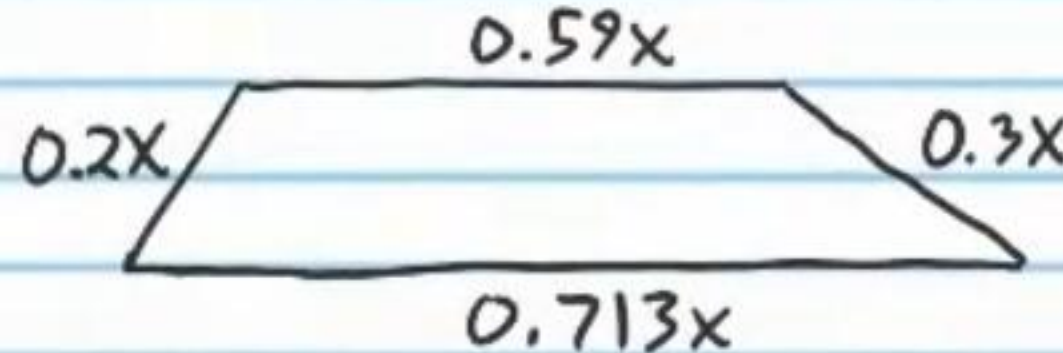
Find an expression for the perimeter of each figure below.

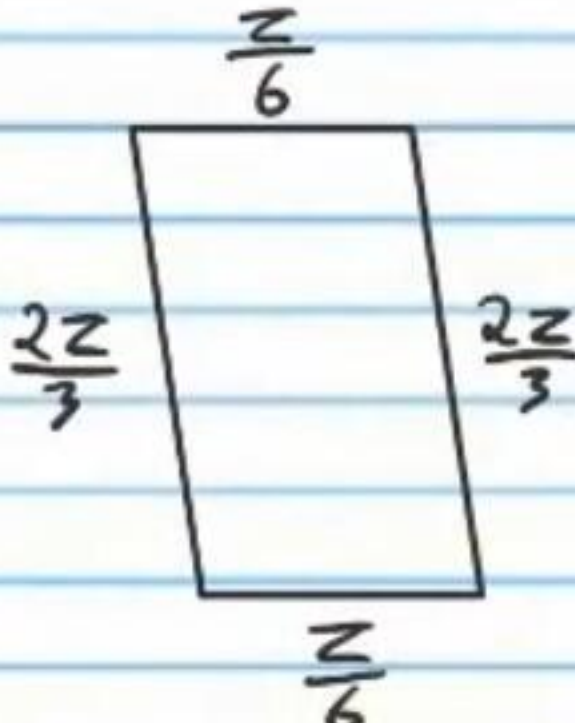
①  $P = 2x + 2x + 5x + 5x = \boxed{14x}$

②  $P = 11.9y + 8.2y + 9y = \boxed{29.1y}$

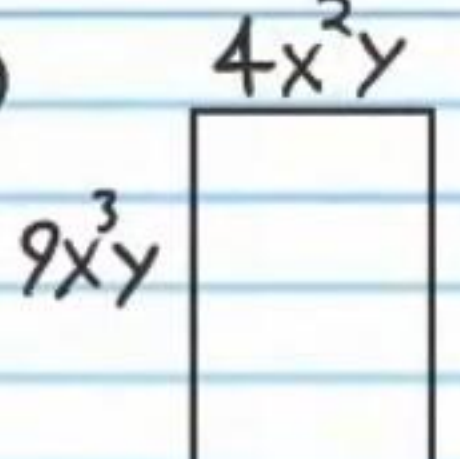
③  $P = \frac{z}{5} + \frac{z}{5} + \frac{2z}{5} + \frac{z}{6} + \frac{z}{2}$
 $P = \frac{4z}{5} + \frac{z}{6} + \frac{z}{2}$
 $P = \frac{24z}{30} + \frac{5z}{30} + \frac{15z}{30} = \frac{44z}{30} = \boxed{\frac{22z}{15}}$

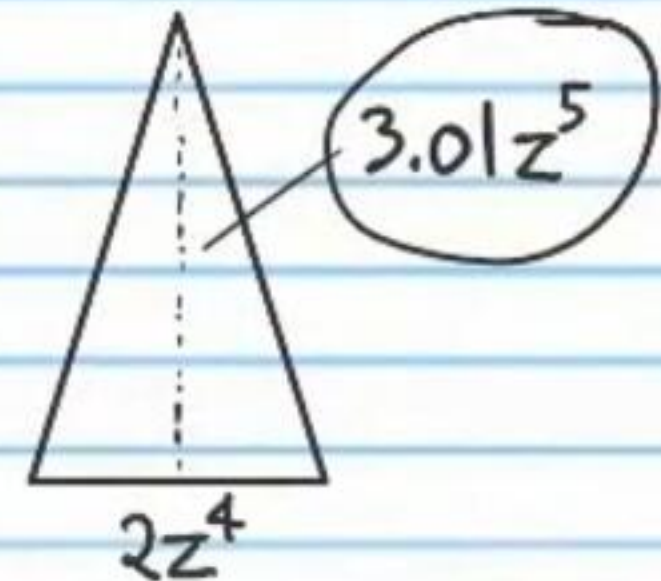
④  $P = 2y + 4y + 5y = \boxed{11y}$

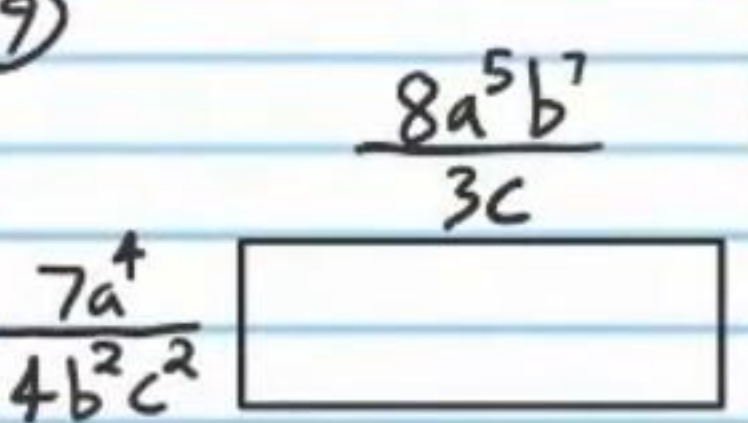
⑤  $P = 0.2x + 0.59x + 0.3x + 0.713x = \boxed{1.803x}$

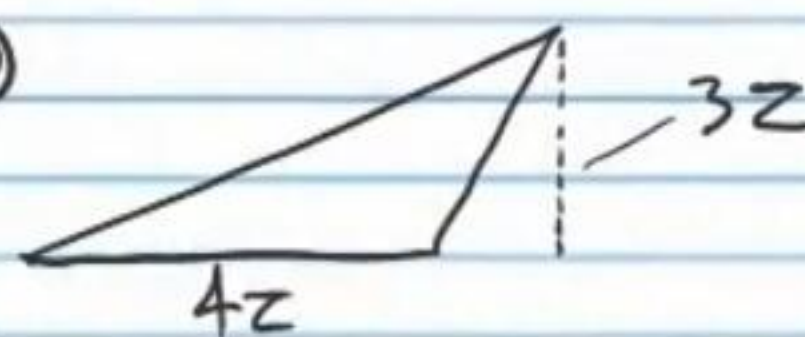
⑥  $P = \frac{z}{6} + \frac{2z}{3} + \frac{2z}{3} + \frac{z}{6}$
 $P = \frac{2z}{6} + \frac{4z}{3}$
 $= \frac{z}{3} + \frac{4z}{3}$
 $= \boxed{\frac{5z}{3}}$

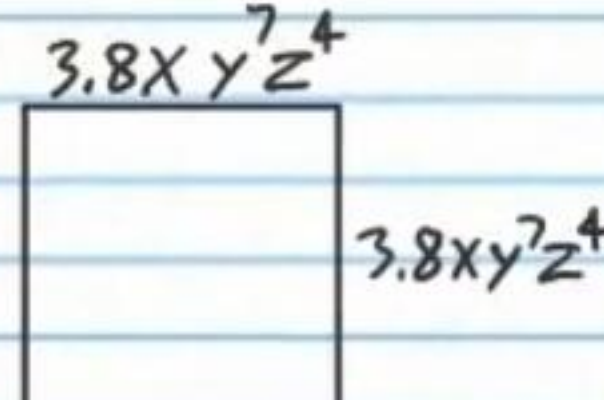
Find an expression for the area of each figure below.

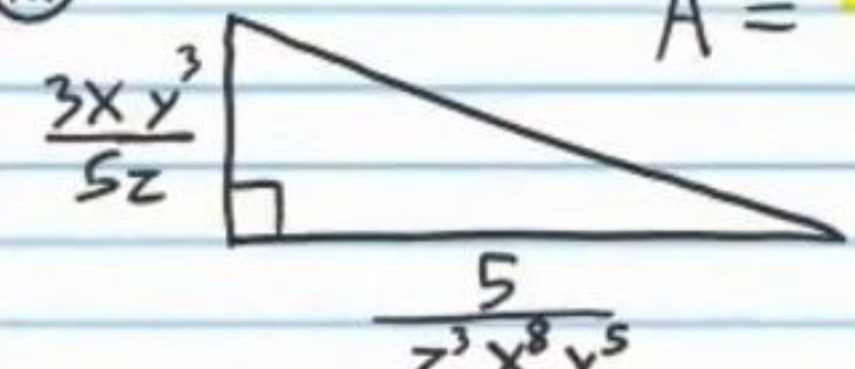
⑦  $A = LW = (9x^3y)(4x^2y)$
 $= \boxed{36x^5y^2}$

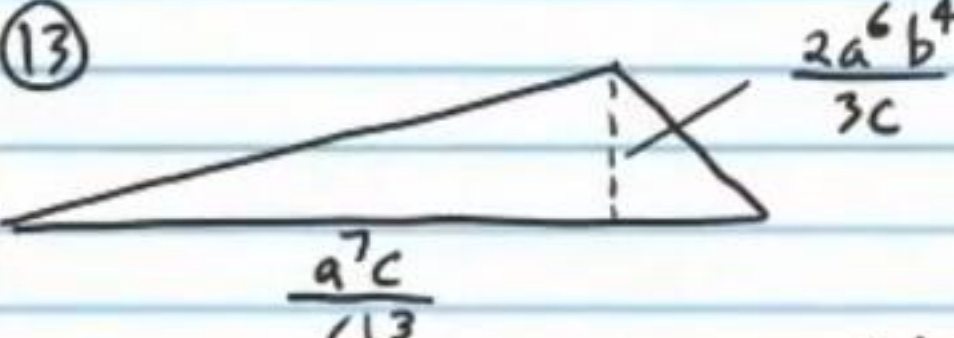
⑧  $A = \frac{bh}{2} = \frac{(2z^4)(3.01z^5)}{2}$
 $= \frac{6.02z^9}{2}$
 $= \boxed{3.01z^9}$

⑨  $A = LW = \left(\frac{7a^4}{4b^2c^2}\right)\left(\frac{8a^5b^7}{3c}\right)$
 $= \frac{7 \cdot 8 a^9 b^7}{4 \cdot 3 b^2 c^3} = \frac{7 \cdot 2 \cdot 4 a^9 b^5}{4 \cdot 3 c^3}$
 $= \boxed{\frac{14a^9b^5}{3c^3}}$

⑩  $A = \frac{bh}{2} = \frac{(4z)(3z)}{2}$
 $= \frac{12z^2}{2}$
 $= \boxed{6z^2}$

⑪  $A = LW = (3.8xy^7z^4)(3.8xy^7z^4)$
 $= \boxed{14.44x^2y^{14}z^8}$

★ ⑫  $A = \frac{bh}{2} = \frac{\left(\frac{3xy^3}{5z}\right)\left(\frac{5}{z^3x^8y^5}\right)}{2}$
 $= \frac{15xy^3}{5z^4x^8y^5} = \frac{3}{z^4x^7y^2} \div \frac{2}{1}$
 $= \left(\frac{3}{z^4x^7y^2}\right)\left(\frac{1}{2}\right)$
 $= \boxed{\frac{3}{2z^4x^7y^2}}$

⑬  $A = \frac{bh}{2} = \frac{\left(\frac{a^7c}{6b^3}\right)\left(\frac{2a^6b^4}{3c}\right)}{2}$
 $= \frac{2a^{13}cb^4}{18b^3c} = \frac{a^{13}b}{9}$
 $= \frac{a^{13}b}{9} \div \frac{2}{1}$
 $= \frac{a^{13}b}{9} \left(\frac{1}{2}\right)$

Geometric Formulas

Perimeter of a Rectangle or Square: $P = 2L + 2W$

Area of a Rectangle or Square: $A = LW$

Area of a Triangle: $A = \frac{bh}{2}$

Volume of a Rectangular Prism: $V = LWH$

Surface Area of a Rectangular Prism: $A = 2LW + 2LH + 2WH$

Area of a Parallelogram: $A = bh$

Perimeter of a Circle (i.e. circumference): $P = 2\pi r$ or πD

Area of a circle: $A = \pi r^2$

Volume of a Sphere: $V = \frac{4\pi r^3}{3}$

Surface Area of a Sphere: $A = 4\pi r^2$

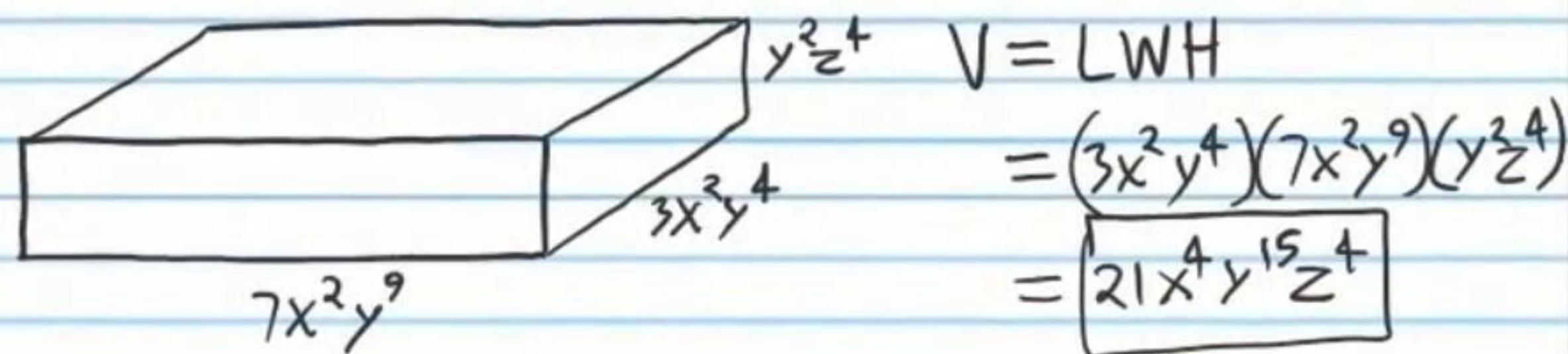
Volume of a Cylinder: $V = \pi r^2 h$

Volume of a Cone: $V = \frac{\pi r^2 h}{3}$

Area of a Trapezoid: $A = \frac{h(b_1 + b_2)}{2}$

Find an expression for each value below.

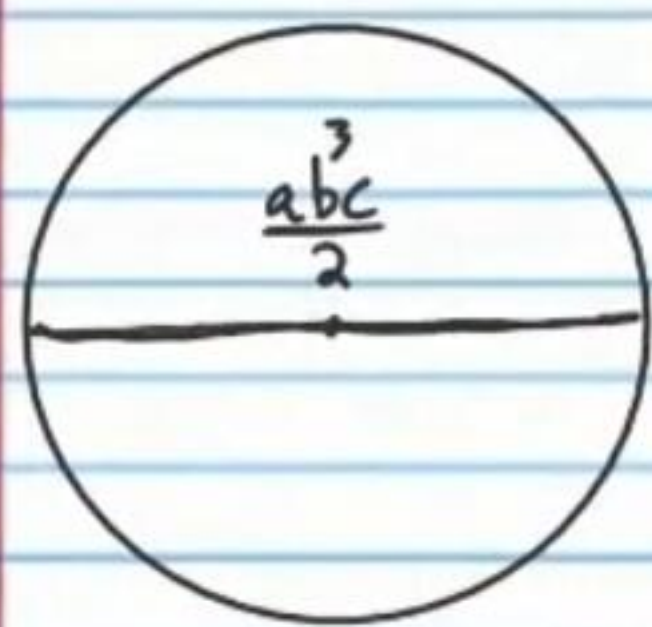
- ⑭ The volume of the rectangular prism below.



- ⑮ The volume of a rectangular prism with length x^3y^3 , width $5xy^2$, and height $2x^2y$.

$$V = LWH = (x^3y^3)(5xy^2)(2x^2y) = \boxed{10x^6y^6}$$

- ⑯ The circumference of the circle below (keep π as a factor).



$$C = \pi D = \pi \left(\frac{abc}{2} \right)$$

$$= \boxed{\frac{abc\pi}{2}}$$

- ⑰ The circumference of the circle below (use 3.14 for π).



$$C = 2\pi r = 2\pi (2.08x^2y^2z^2)$$

$$= \boxed{4.16\pi x^2y^2z^2}$$

- ⑱ The area of a circle with radius $3b^2c$.

$$A = \pi r^2 = \pi (3b^2c)^2 = \pi (3b^2c)(3b^2c)$$

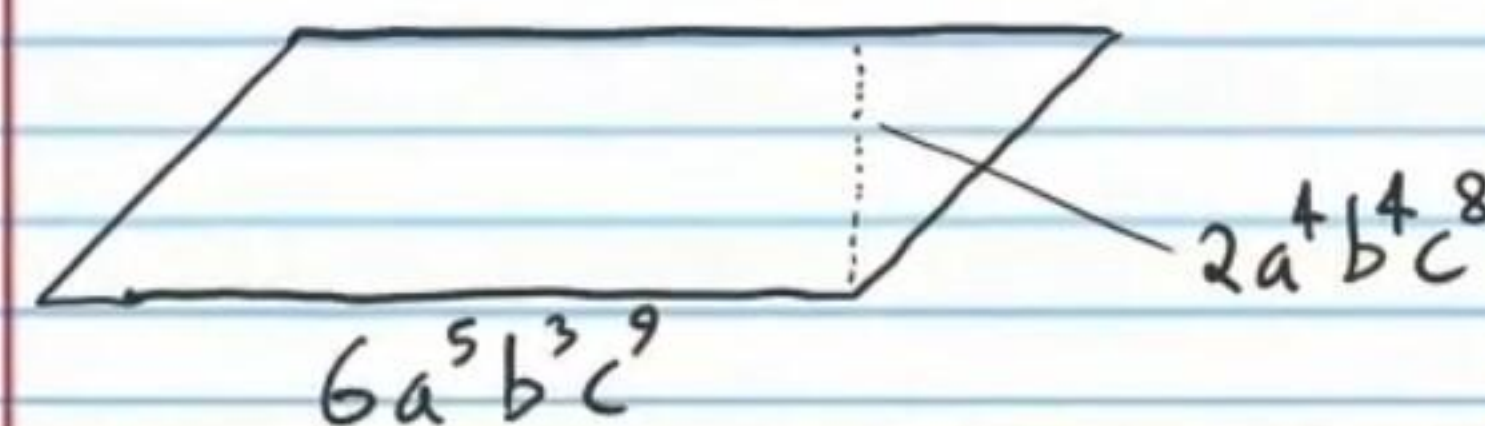
$$= \boxed{9\pi b^4c^2}$$

- ⑲ The area of a circle with radius $9zy$.

$$A = \pi r^2 = \pi (9zy)^2 = \pi (9zy)(9zy)$$

$$= \boxed{81\pi z^2y^2}$$

- ⑳ The area of the parallelogram below.

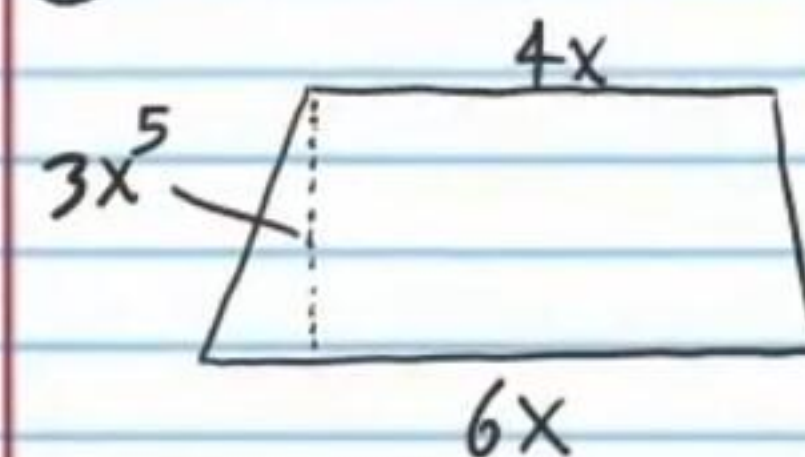


$$A = bh = (6a^5b^3c^9)(2a^4b^4c^8) = \boxed{12a^9b^7c^{17}}$$

- ㉑ The area of a parallelogram with base $\frac{2x}{3}$ and height $10x^4y$.

$$A = bh = \left(\frac{2x}{3} \right) (10x^4y) = \left(\frac{2x}{3} \right) \left(\frac{10x^4y}{1} \right) = \boxed{\frac{20x^5y}{3}}$$

- ㉒ The area of the trapezoid below.

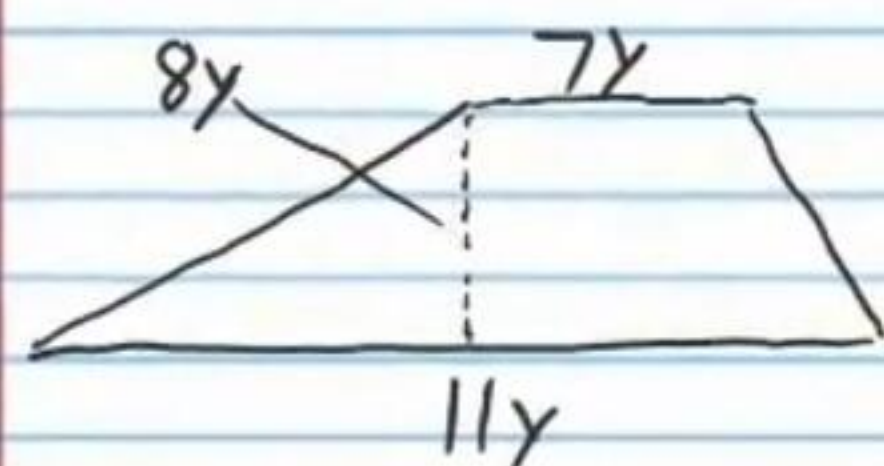


$$A = \frac{h(b_1 + b_2)}{2} = \frac{3x^5(4x + 6x)}{2}$$

$$= \frac{3x^5(10x)}{2}$$

$$= \frac{30x^6}{2} = \boxed{15x^6}$$

②③ The area of the trapezoid below.

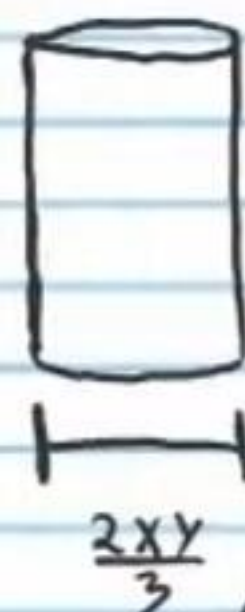


$$A = \frac{h(b_1 + b_2)}{2} = \frac{8y(11y + 7y)}{2}$$

$$= \frac{8y(18y)}{2}$$

$$= \frac{144y^2}{2} = \boxed{72y^2}$$

②④ The volume of the cylinder below (keep π as a factor).



$$r = \frac{D}{2} = \frac{\frac{2xy}{3}}{2} = \frac{2xy}{3} \div 2 = \frac{2xy}{3} \left(\frac{1}{2}\right) = \frac{xy}{3}$$

$$V = \pi r^2 h$$

$$V = \pi \left(\frac{xy}{3}\right)^2 \left(\frac{5x^2y^3}{3}\right)$$

$$V = \pi \left(\frac{xy}{3}\right) \left(\frac{xy}{3}\right) \left(\frac{5x^2y^3}{3}\right) = \boxed{\frac{5\pi x^4 y^5}{27}}$$

②⑤ The volume of a cylinder with radius $5.8z$ and height $3.6z^5$.

$$V = \pi r^2 h = \pi (5.8z)^2 (3.6z^5)$$

$$= \pi (5.8z^3)(5.8z^3)(3.6z^5) = \boxed{121.104\pi z^{11}}$$

②⑥ The surface area of the sphere below (keep π as a factor).



$$A = 4\pi r^2 = 4\pi (2.1yz)^2$$

$$= 4\pi (2.1yz)(2.1yz)$$

$$= \boxed{17.64\pi z^2 y^2}$$

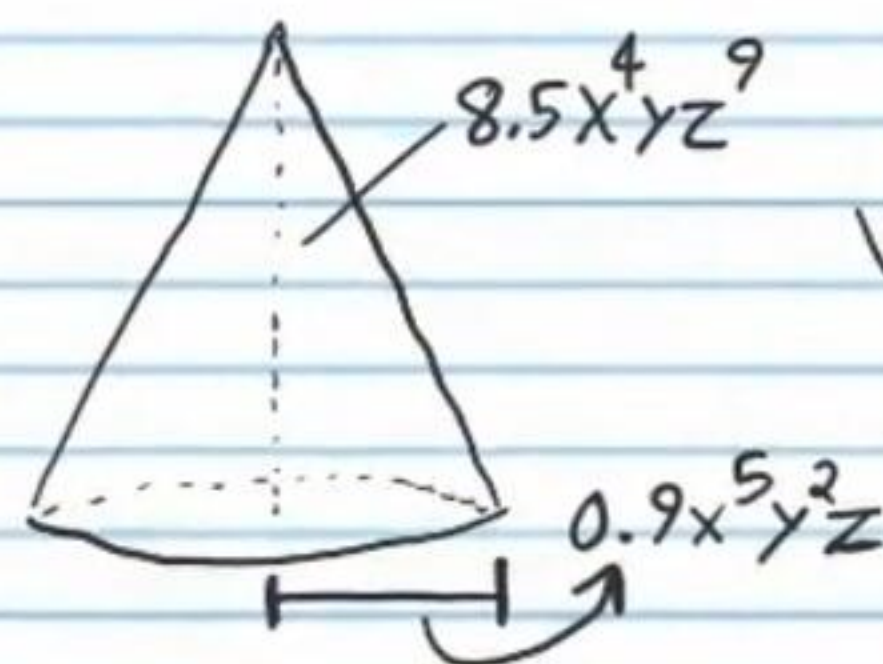
②⑦ The surface area of a sphere with diameter $\frac{3xy^5}{4}$.

$$r = \frac{D}{2} = \frac{\frac{3xy^5}{4}}{2} = \frac{3xy^5}{4} \div 2 = \frac{3xy^5}{4} \left(\frac{1}{2}\right) = \frac{3xy^5}{8}$$

$$A = 4\pi r^2 = 4\pi \left(\frac{3xy^5}{8}\right)^2 = 4\pi \left(\frac{3xy^5}{8}\right) \left(\frac{3xy^5}{8}\right)$$

$$= 4\pi \left(\frac{9x^2 y^{10}}{64}\right) = \frac{4\pi (9x^2 y^{10})}{64}$$

②⑧ The volume of the cone below.



$$V = \frac{\pi r^2 h}{3}$$

$$= \frac{\pi (0.9x^5y^2z)^2 (8.5x^4yz^9)}{3}$$

$$= \frac{6.885\pi x^{14} y^5 z^{11}}{3} = \boxed{2.295\pi x^{14} y^5 z^{11}}$$

②⑨ The volume of a cone with radius $3y^2z^{10}$ and height $9y^3z^4$.

$$V = \frac{\pi r^2 h}{3} = \frac{\pi (3y^2z^{10})^2 (9y^3z^4)}{3} = \pi (3y^2z^{10})(3y^2z^{10})(9y^3z^4)$$

$$= \boxed{27\pi y^7 z^{24}}$$

Find the average (i.e. mean) of the following data sets.

③⑩ $2x, 5x, 4x, 5x$ Mean = $\frac{2x + 5x + 4x + 5x}{4}$

$$= \frac{16x}{4} = \boxed{4x}$$

③⑪ $-3y, -y, 2y, -4y, -y$ Mean = $\frac{-3y - y + 2y - 4y - y}{5}$

$$= \boxed{-\frac{7y}{5}}$$

32) $10z, 11z, 9z, 5z, 9z, 10z$

$$\text{Mean} = \frac{10z + 11z + 9z + 5z + 9z + 10z}{6}$$

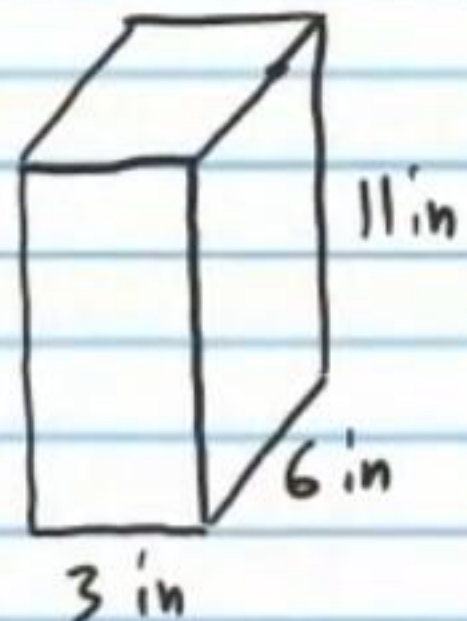
$$= \frac{54z}{6} = \boxed{9z}$$

33) $x, 2x, x, 2x, 3x$

$$\text{Mean} = \frac{x + 2x + x + 2x + 3x}{5} = \boxed{\frac{9x}{5}}$$

Evaluating Algebraic Expressions

34) Find the surface area of the rectangular prism below.



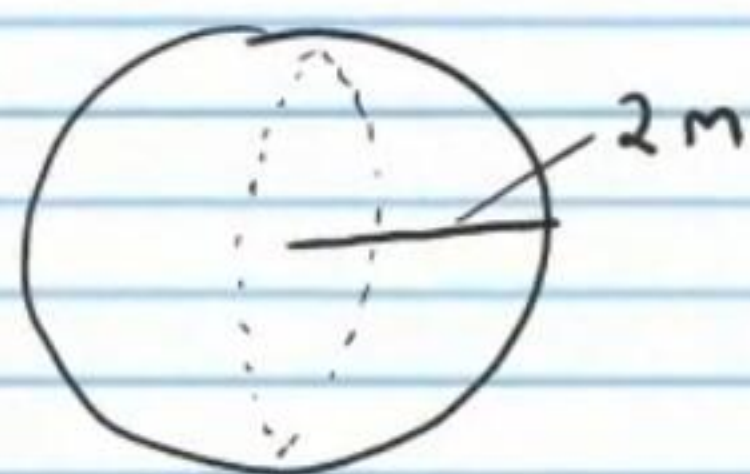
$$A = 2LH + 2LW + 2WH$$

$$A = 2(6)(11) + 2(6)(3) + 2(3)(11)$$

$$A = 132 + 36 + 66$$

$$\boxed{A = 234 \text{ in}^2}$$

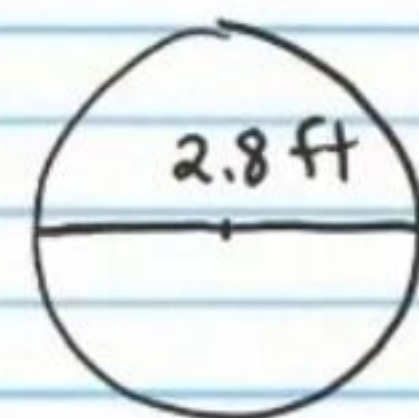
35) Find the volume of the sphere below (keep π as a factor).



$$V = \frac{4\pi r^3}{3}$$

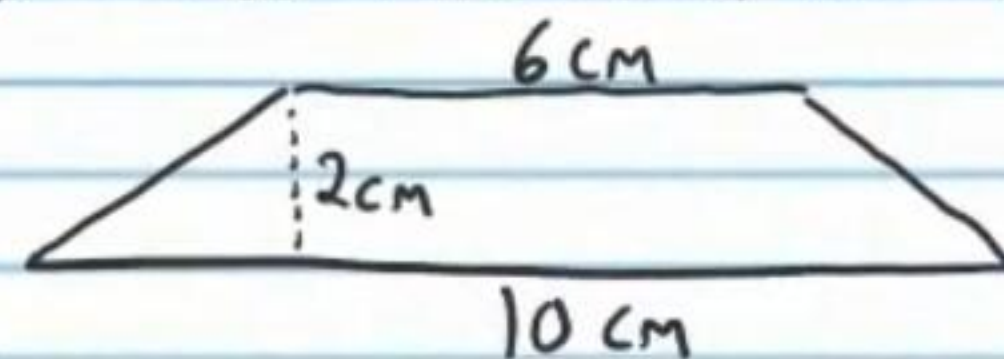
$$= \frac{4\pi(2)^3}{3} = \frac{4\pi(8)}{3} = \boxed{\frac{32\pi}{3} \text{ m}^3}$$

36) Find the circumference of the circle below (keep π as a factor).



$$C = D\pi = \boxed{2.8\pi \text{ ft}}$$

37) Find the area of the trapezoid below.



$$A = \frac{h(b_1 + b_2)}{2}$$

$$= \frac{2(6 + 10)}{2} = \frac{2(16)}{2} = \frac{32}{2} = \boxed{16 \text{ cm}^2}$$

38) Find the surface area of a sphere with diameter 10 m (keep π as a factor).

$$r = 10 \div 2 = 5$$

$$A = 4\pi r^2 = 4\pi(5)^2 = 4\pi(25) = \boxed{100\pi \text{ m}^2}$$

39) Find the area of a parallelogram with base $\frac{2}{3}$ in and height $\frac{1}{6}$ in.

$$A = bh = \left(\frac{2}{3}\right)\left(\frac{1}{6}\right) = \frac{2}{18} = \boxed{\frac{1}{9} \text{ in}^2}$$

40) Find the volume of a cone with radius 5 ft and height 9 ft.

$$V = \frac{\pi r^2 h}{3} = \frac{\pi(5)^2(9)}{3} = \frac{\pi(25)(9)}{3} = \frac{\pi(25)(3)(3)}{3} = \boxed{75\pi \text{ ft}^3}$$

41) Find the area of a circle with diameter 556 mi.

$$A = \pi r^2 = \pi(278)^2 = \boxed{77,284\pi \text{ mi}^2}$$

$$r = D/2 = 556/2 = 278$$

④② Evaluate the answer for problem #1 if $x=3$.

$$P = 14x = 14(3) = \boxed{42}$$

④③ Evaluate the answer for problem #7 if $x=1$ and $y=2$.

$$A = 36x^5y^2 = 36(1)^5(2)^2 = 36(1)(4) \\ = 36(4) \\ = \boxed{144}$$

④④ Evaluate the answer for problem #12 if $x=1$, $y=1$, and $z=1$.

$$A = \frac{3}{2z^4x^7y^2} = \frac{3}{2(1)^4(1)^7(1)^2} = \frac{3}{2(1)(1)(1)} = \boxed{\frac{3}{2}}$$

④⑤ Evaluate the answer for problem #8 if $z=2.1$.

$$A = 3.01z^9 = 3.01(2.1)^9 \approx \boxed{2,390.78}$$

④⑥ Use the simple interest formula below to find the amount of money in a bank account if $P=200$, $r=0.02$, and $t=10$.

$$A = P + Prt = 200 + 200(0.02)(10) \\ = 200 + 4(10) \\ = 200 + 40 = \boxed{240}$$

④⑦ Use the percent increase formula below to find Albert's new height if $F=48$ and $r=0.05$.

$$N = F + Fr = 48 + 48(0.05) \\ = 48 + 2.4 \\ = \boxed{50.4}$$